

Cross-modal Representation Learning for Diffusion-generated Image Detection

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Abstract

The astonishing proficiency and unprecedented level of realism of diffusion models in creating and manipulating images have undoubtedly drawn concerns. Many methods have been proposed to detect generated images. Typically, they usually take RGB images as input, and use backbones like ResNet, CLIP visual encoder to extract features. Even though these backbones are capable to detect fake images, they are mainly designed to extract the high-level semantic information, rather than inherently designed for fake image detection. To this end, in this paper, we want to optimize the embedding space tailored for detecting fake images via representation learning. We notice that Neighboring Pixel Relationships (NPR) is capable to capture the intrinsic forgery clues, which means that NPR may be a good input to perform representation learning that aims at learning the embedding space tailored for detecting fake images. Therefore, we leverage features from both RGB modality and NPR modality to perform two proposed representation learning methods, Cross-Modal Contrastive Learning (CMCL) and Cross-Modal Mutual Distillation (CMMD), in order to learn the forgery-aware embedding space. The CMCL boosts the discrimination of features between real and fake images, while the CMMD simultaneously transfers the learned knowledge between two modalities, being able to learn compact features within the intra-class. CMCL and CMMD work collaboratively so that each modality learns a more comprehensive forgery-aware representation to distinguish real and fake images. Extensive experiments on GenImage, DRCT-2M, and Co-Spy-Bench datasets show that our method achieves state-of-the-art results.

1. Introduction

In recent years, diffusion-based image generation technologies have shown impressive results in generating realistic images. These technologies have provided efficient content

editing and generation tools for applications, such as digital creation, commercial advertising, and social entertainment. However, rapid advancements in synthetic technologies have raised significant concerns regarding their potential implications, such as information security and personal privacy. Therefore, there is an urgent need to develop technologies for detecting generated images.

Numerous studies have explored various aspects of diffusion models, including controllable generation [52], optimizing generative architectures [24, 39], accelerating the generation process [12, 41], and much more. The wide diversity of image generative models has raised a considerable challenge to the generalizability of detection methods. It demands that generated image detectors should be able to identify images produced by not only the known generative models, but also the newly developed models that have not been involved in the training of detectors.

Recent advancements aimed at enhancing this generalization ability include the refinement of detection algorithms [9, 21, 23, 26, 32, 50] and the development of pre-trained models [4, 10, 33, 40, 42, 51]. Typically, these methods usually take RGB images as input, and use backbones like ResNet, CLIP visual encoder to extract features for fake image detection. However, even though these backbones are capable to detect fake images, they are mainly designed to extract the high-level semantic information, rather than inherently designed for fake images detection. CoDE [2] make a simple attempt towards embedding space tailored for generated image detection, verifying the effectiveness of building forgery-aware embedding space. However, CoDE simply utilizes conventional contrastive learning with only RGB as input. In the paper, we explore a more effective way to optimize the embedding space specially tailored for detecting fake images via representation learning, i.e., a forgery-aware embedding for distinguishing fake and real images.

Previous works have shown that RGB is a good input to detect fake images. However, a conspicuous gap remains in the lack of source-invariant representation exploited from the generator pipeline for generated image detection. [44]

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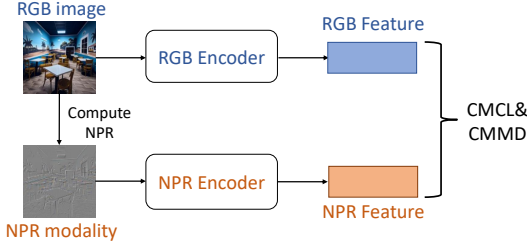


Figure 1. Overview of the proposed detector with CMCL and CMMD.

focuses on achieving source-invariant fake image detection by rethinking artifacts stemming from the up-sampling component of common generation models. They propose a simple but effective artifact representation, termed Neighboring Pixel Relationships (NPR), aimed at achieving generalized fake image detection. NPR is capable to capture the intrinsic forgery clues, and shows good generalization capabilities across different generation models. Therefore, NPR may be a good input to perform representation learning that aims at learning the forgery-aware embedding space.

To this end, in this paper, as shown in Figure 1, we leverage features from both RGB modality and NPR modality to perform two proposed representation learning methods, Cross-Modal Contrastive Learning (CMCL) and Cross-Modal Mutual Distillation (CMMD), in order to learn the forgery-aware embedding space. Specifically, the CMCL aims at increasing the inter-class separability (real class vs fake class) by contrastive learning, while the CMMD is able to learn compact features within the intra-class (real class or fake class) by modeling the relationships within the same class. The CMCL and CMMD are aimed at learning a comprehensive forgery-aware embedding space tailored for fake image detection, rather than the high-level semantic features.

Intuitively, taking the RGB feature of a real image as an example, the RGB features and the corresponding NPR features of the same real image should be closer than the RGB feature of the real image and NPR feature of other fake images. Therefore, we propose CMCL to optimize the RGB embedding space tailored for distinguishing real and generated images. Specifically, as shown in Figure 2 (a), the RGB feature and the NPR feature from the same real image constitute the positive sample pair, while the RGB feature of the real image and the NPR features from other fake images constitute the negative sample pairs. CMCL uses contrastive loss to optimize the positive sample pair and negative sample pairs, in order to boost the discrimination of the RGB feature between real and fake images. Similarly, CMCL can boost the discrimination of the NPR feature between real and fake images.

The above CMCL utilizes the RGB modality and the

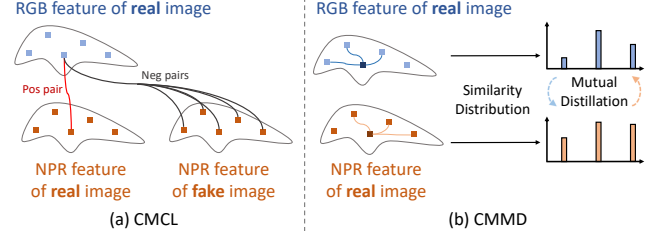


Figure 2. The overview of the proposed CMCL and CMMD with the RGB feature of a real image as an example.

NPR modality to boost the discrimination of the RGB feature and NPR feature **between** real images and fake images. However, the CMCL can not model the relationships **within** the real images (or fake images). Taking real images as an example, the knowledge in each modality is not been fully explored. Therefore, we propose CMMD to utilize the knowledge in each modality. Specifically, as shown in Figure 2 (b), the neighboring similarity distribution is first extracted in each modality. It describes the relationship of the sample embedding with respect to its nearest neighbors in the customized feature space. Based on the relational information, bidirectional knowledge distillation between the two modalities is performed via explicit cross-modal consistency constraints.

The CMCL utilizes the cross-modal feature between real images and fake images to boost the discrimination of features, while the CMMD utilizes the cross-modal feature within the real images (or fake images) to transfer learned knowledge between RGB and NPR modalities. CMCL and CMMD run synchronously and cooperatively so that each modality learns a more comprehensive forgery-aware representation.

Our contributions are summarized as follows:

1. To optimize a forgery-aware embedding space tailored for fake image detection, we leverage the RGB features and NPR features to perform the proposed Cross-Modal Contrastive Learning (CMCL). CMCL can boost the discrimination of the RGB features and NPR features between real and fake images. (Note that NPR is **not** claimed as our contribution, since it has been proposed in [44])
2. To fully utilize the neighboring similarity distribution in each modality of real images (or fake images), we propose Cross-Modal Mutual Distillation (CMMD) to transfer the learned knowledge between RGB and NPR modalities. The proposed CMMD can learn more compact embedding within the intra-class.
3. We propose a Strong Diffusion-generated Image Detector (SDID) by taking advantage of two different but complementary representation learning methods CMCL and CMMD. Extensive experiments on three benchmarks,

GenImage, DRCT-2M, and Co-Spy-Bench, verify the effectiveness of the proposed SDID, and show that our method achieves state-of-the-art results on diffusion-generated image detection.

2. Related Work

2.1. AI-Generated Image Detection

Early approaches, such as [29–31] demonstrated limited generalizability as generative models advanced. CNNSpot [47] made significant progress by showing that a classifier trained only on ProGAN outputs could generalize to unseen GAN architectures. Subsequent work FreqFD [14] revealed frequency-domain artifacts in GAN-generated images.

Recent advances have explored more sophisticated detection paradigms to enhance generalization. DIRE [3, 28, 38, 48] proposed a novel approach using differences between images and their reconstructions from a pretrained reconstruction model as discriminative features. PatchCraft [53] achieved state-of-the-art generalization by analyzing inter-pixel correlation discrepancies between rich-texture and poor-texture patches. Several studies have successfully leveraged large pretrained models: UnivFD [33] combined CLIP features with nearest-neighbor classification, FatFormer [23] incorporated a forgery-aware adapter into CLIP to integrate local manipulation traces, De-fake [40] employed CLIP-based multimodal fusion, CLIPMoLE [26] adopted a combination of shared and separate LoRAs within an MoE-based structure in deeper ViT blocks, and AIDE [49] extended this approach by using multiple expert modules to simultaneously capture visual artifacts and noise patterns. Co-Spy [9] enhances and integrates existing CLIP-based semantic features and artifact features extracted by VAE. CoDE [2] only utilizes contrastive learning with RGB as inputs to train an embedding designed for generated image detection. The above methods usually take the RGB image as input, while [44] proposes a simple but effective artifact representation, termed Neighboring Pixel Relationships (NPR), to achieve source-invariant fake image detection. NPR is better than RGB to capture the intrinsic forgery clues. [55] is the first work using both RGB and NPR modalities. However, [55] focuses on leveraging MLLM with RGB and NPR features for the explanations of fake images. In this paper, we explore leveraging features from both RGB and NPR modality to optimize a forgery-aware embedding space via representation learning.

2.2. Representation Learning

Contrastive learning [6, 17, 19] aims to learn feature representation via instance discrimination. It pulls positive pairs closer and pushes negative pairs away. Since no labels are available during self-supervised contrastive learning, two different augmented versions of the same sample are treated

as a positive pair, and samples from different instances are considered to be negative. In MoCo [17] and MoCo v2 [7], the negative samples are taken from previous batches and stored in a queue-based memory bank. In contrast, SimCLR [6] and MoCo v3 [8] rely on a larger batch to provide enough negative samples. Pairwise relationships has been shown to be useful information in relational knowledge distillation [34, 36, 45]. In PKT [35], CompRes [1], and SEED [13], knowledge distillation is performed on the relationships of each sample with respect to a set of anchors, which models the structural information of the data. In this paper, we utilize the cross-modal (RGB and NPR) features to explore cross-modal representation learning to learn forgery-aware embedding space tailored for fake image detection.

3. Method

3.1. Framework Overview

Figure 3 illustrates the overall pipeline of our framework. We only use the real images as examples to illustrate the proposed CMCL and CMMD for convenience. The CMCL and CMMD for fake images are similar to those of real images, since the architecture is symmetric.

Firstly, the NPR modality of real image I_{real}^{NPR} is computed by the RGB modality of real image I_{real}^{RGB} . Then, I_{real}^{RGB} and I_{real}^{NPR} are passed through the RGB image encoder E^{RGB} and the NPR image encoder E^{NPR} to extract the RGB feature F_{real}^{RGB} and the NPR feature F_{real}^{NPR} , respectively. The F_{real}^{RGB} and F_{real}^{NPR} are stored in queue Q_{real}^{RGB} and Q_{real}^{NPR} , respectively. Both Q_{real}^{RGB} and Q_{real}^{NPR} are defined on the fly with the current batch enqueued and the oldest batch dequeued, during training. Similarly, we can get the I_{fake}^{RGB} , I_{fake}^{NPR} , F_{fake}^{RGB} , F_{fake}^{NPR} , Q_{fake}^{RGB} , Q_{fake}^{NPR} belonging to the fake images.

The F_{real}^{RGB} , F_{real}^{NPR} , and Q_{fake}^{NPR} are used to perform Cross-Modal Contrastive Learning (CMCL). Similarly, F_{real}^{NPR} , F_{real}^{RGB} , and Q_{fake}^{RGB} are used to perform CMCL. These two CMCL aim to boost the discrimination of F_{real}^{RGB} and F_{real}^{NPR} between real images and fake images. However, the CMCL can only boost the feature discrimination between real and fake images, and cannot model the relationships within the real images. Therefore, we use F_{real}^{RGB} , F_{real}^{NPR} , Q_{real}^{RGB} , and Q_{real}^{NPR} to perform Cross-Modal Mutual Distillation (CMMD) in order to transfers the learned knowledge between RGB and NPR modalities. Finally, we concatenate the feature F_{real}^{RGB} and F_{real}^{NPR} , which are already enhanced by the proposed CMCL and CMMD, to predict the label of the real image with cross-entropy loss L_{cls} . After training, only the E^{RGB} and E^{NPR} are reserved for generated image detection. The features from E^{RGB} and E^{NPR} are concatenated to predict whether the input image is real or fake.

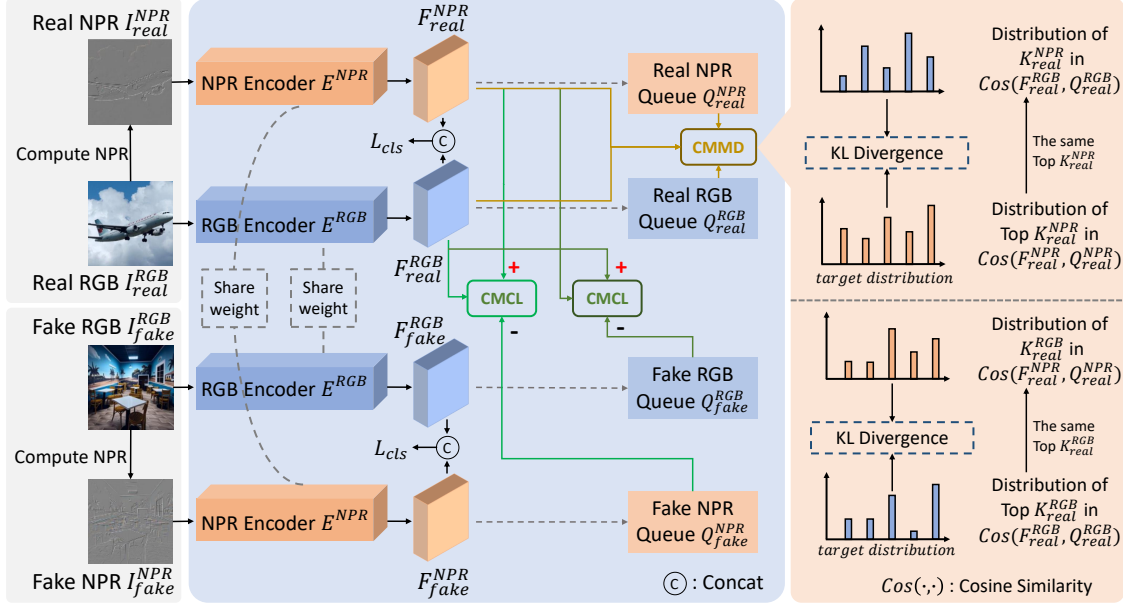


Figure 3. The overall pipeline of SDID. We only use the real images as examples to illustrate the proposed CMCL and CMMD for convenience. The CMCL and CMMD for fake images are similar to those of real images, since the architecture is symmetric.

3.2. Neighboring Pixel Relationships

Neighboring Pixel Relationships (NPR) was first proposed in [44] to develop a generalizable artifacts representation for generated image detection. Their findings reveal that the up-sampling operator can produce generalized forgery artifacts. Existing works typically consider its influence on the whole image in the frequency domain. In contrast, they explore the trace of the up-sampling layer from the local image pixels, and present a simple but effective artifact representation NPR to achieve generalized generated image detection. Please refer to [44] for detailed motivation and theory.

The NPR of an image is computed from its RGB modality. Specifically, the RGB image is divided into non-overlapping patches with patch sizes $l \times l$ ($l = 2$ as suggested by [44]). Each pixel in a patch subtracts the value of the top-left pixel per channel in the same patch. We can get the NPR modality of the image after preprocessing all patches. Please refer to [44] for more technical details.

3.3. Cross-modal Contrastive Learning

Intuitively, taking the RGB feature F_{real}^{RGB} of a real image as an example, the F_{real}^{RGB} and the corresponding NPR feature F_{real}^{NPR} of the same real image should be closer than F_{real}^{RGB} of the real image and the NPR features of other fake images in Q_{fake}^{NPR} . Therefore, we propose Cross-Modal Contrastive Learning (CMCL) to boost the discrimination of RGB features between real images and fake images, increasing the inter-class separability (real class vs fake class)

We use InfoNCE [6, 17] loss as contrastive loss. InfoNCE loss seeks to simultaneously pull close positive views from the same sample and push away negative samples. The loss function of InfoNCE loss for the RGB feature F_{real}^{RGB} of the real image is :

$$L_{CMCL}(F_{real}^{RGB}; F_{real}^{NPR}; Q_{fake}^{NPR}) = -\log \frac{\exp(F_{real}^{RGB} \cdot F_{real}^{NPR} / \tau)}{\exp(F_{real}^{RGB} \cdot F_{real}^{NPR} / \tau) + \sum_{q_i \in Q_{fake}^{NPR}} \exp(F_{real}^{RGB} \cdot q_i / \tau)} \quad (1)$$

where $\log$ denotes the log operation, and τ is the temperature constant, which is set to 0.07. N_Q is the number of features stored in the queue Q_{fake}^{NPR} . The $F_{real}^{RGB} \cdot F_{real}^{NPR}$ of the same real image constitute the positive pair, and the $F_{real}^{RGB} \cdot q_i$ ($q_i \in Q_{fake}^{NPR}$) constitute the negative pairs. Due to the limitation of GPU memory, the Q_{fake}^{NPR} are defined on the fly with the current batch enqueued and the oldest batch dequeued, during training.

The final CMCL loss is defined as:

$$L_{CMCL} = L_{CMCL}(F_{real}^{RGB}; F_{real}^{NPR}; Q_{fake}^{NPR}) + L_{CMCL}(F_{real}^{NPR}; F_{real}^{RGB}; Q_{fake}^{RGB}) + L_{CMCL}(F_{fake}^{RGB}; F_{fake}^{NPR}; Q_{real}^{NPR}) + L_{CMCL}(F_{fake}^{NPR}; F_{fake}^{RGB}; Q_{real}^{RGB}) \quad (2)$$

With the proposed CMCL, we can boost the discrimination of both RGB and NPR features between real images and fake images.

3.4. Cross-modal Mutual Distillation

With the proposed CMCL and cross-entropy loss L_{cls} , the model could learn a forgery-aware embedding space tailored for fake image detection. However, it would be unreasonable to just ignore potential information interaction between different modalities of samples within the same class (such as real class), since they belong to the same class. In other words, taking real images as an example, the knowledge in each modality of the real class has not been fully explored. Therefore, we propose Cross-Modal Mutual Distillation (CMMD) to model the learned knowledge and transfer the biased knowledge between complementary modalities. Note that we limit the mutual distillation within the real images (or fake images), which means that there is no distillation between real images and fake images. CMMD is able to learn compact forgery-aware features within the intra-class (real class or fake class).

To perform knowledge distillation between modalities, we first need to model the knowledge learned in each modality in a proper way. It needs to take advantage of the existing cross-modal contrastive learning framework to avoid introducing excessive computational overhead. We utilize the pairwise relationship between samples for modality-specific knowledge modeling. Specifically, given an embedding z and a set of anchors $A = \{a_i\}_{i=1,2,\dots,K}$, we compute the cosine similarities between them as $\cos(z, a_i) = z \cdot a_i, i = 1, 2, \dots, K$. The queues in contrastive learning store a handful of embeddings of the same modality. We can easily obtain the required anchors without additional model inference. We select the top K nearest neighbors of z as the anchors. The resulting pairwise similarities are further converted into probability distributions with the temperature constant τ :

$$p_i(z, a_i) = \frac{\exp(\cos(z, a_i)/\tau)}{\sum_{j=1}^K \exp(\cos(z, a_j)/\tau)}, i = 1, 2, \dots, K. \quad (3)$$

The obtained $p(z, A) = \{p_i(z, a_i)\}_{i=1,2,\dots,K}$ describes the distribution characteristic around the embedding z in the customized feature space of each modality.

Based on the aforementioned probability distributions, an intuitive way to perform knowledge distillation would be to directly establish consistency constraints between RGB and NPR modalities. Different from previous knowledge distillation approaches that transfer the knowledge of a fixed and well-trained teacher model to the student model, in our approach, the knowledge is continuously updated during training, and each modality acts as both student and teacher. Knowledge distillation from modality RGB to modality NPR for real images is performed by minimizing the fol-

lowing KL divergence:

$$\begin{aligned} L_{CMMD}(F_{real}^{RGB}; Q_{real}^{RGB}; F_{real}^{NPR}; Q_{real}^{NPR}) \\ = \text{KL}(p(F_{real}^{RGB}, Q_{real}^{RGB}) || p(F_{real}^{NPR}, Q_{real}^{NPR})) \\ = \sum_{i=1}^K p_i(F_{real}^{RGB}, (q_{real}^{RGB})_i) \cdot \log \frac{p_i(F_{real}^{RGB}, (q_{real}^{RGB})_i)}{p_i(F_{real}^{NPR}, (q_{real}^{NPR})_i)}. \end{aligned} \quad (4)$$

where $(q_{real}^{RGB})_i \in Q_{real}^{RGB}$ and $(q_{real}^{NPR})_i \in Q_{real}^{NPR}$. Since the knowledge distillation works bidirectionally, knowledge distillation from modality NPR to modality RGB is defined as $L_{CMMD}(F_{real}^{NPR}; Q_{real}^{NPR}; F_{real}^{RGB}; Q_{real}^{RGB})$.

Similarly, we can get the distillation loss for fake images. Therefore, the overall CMMD loss is defined as:

$$\begin{aligned} L_{CMMD} = & L_{CMMD}(F_{real}^{RGB}; Q_{real}^{RGB}; F_{real}^{NPR}; Q_{real}^{NPR}) \\ & + L_{CMMD}(F_{real}^{NPR}; Q_{real}^{NPR}; F_{real}^{RGB}; Q_{real}^{RGB}) \\ & + L_{CMMD}(F_{fake}^{RGB}; Q_{fake}^{RGB}; F_{fake}^{NPR}; Q_{fake}^{NPR}) \\ & + L_{CMMD}(F_{fake}^{NPR}; Q_{fake}^{NPR}; F_{fake}^{RGB}; Q_{fake}^{RGB}) \end{aligned} \quad (5)$$

3.5. Training Loss

The features from E^{RGB} and E^{NPR} are concatenated to predict whether the input image is real or fake. The corresponding classification loss L_{cls} is the cross-entropy loss. The final loss function to train the proposed SDID is the combination of L_{cls} , L_{CMCL} , and L_{CMMD} :

$$L = L_{cls} + \lambda_1 L_{CMCL} + \lambda_2 L_{CMMD} \quad (6)$$

where λ_1 and the λ_2 is used to balance the loss among L_{cls} , L_{CMCL} , and L_{CMMD} . After training, only the E^{RGB} and E^{NPR} are reserved for generated image detection.

4. Experiment

4.1. Datasets and Implementation Details

Experiments are conducted on three benchmarks: GenImage [56], DRCT-2M [4], and Co-Spy-Bench [9]. The evaluation metric is measured in accuracy, with a decision threshold of 0.5.

The RGB encoder network is the pre-trained Dinov2-ViT-L-14 with LoRA [18] to finetune. The NPR encoder is the ResNet-101 [16] pretrained on ImageNet [11]. The length N_Q of each queue used in CMCL is set to 2048, and the Top K nearest neighbors in Equation 3 is set to 128. The loss weight λ_1 and λ_2 are experimentally set to 0.1.

4.2. Main Results

We compare our method with other methods on the GenImage, DRCT-2M, and Co-Spy-Bench datasets, respectively, as shown in Table 1, 2, 3. Table 1 shows that the proposed SDID outperforms all other methods by at least 2 points.

Method	Midjourney	SDv1.4	SDv1.5	ADM	GLIDE	Wukong	VQDM	BigGAN	Avg.
CNNSpot [47]	84.9	99.8	99.7	53.4	53.8	99.6	55.5	49.9	74.6
F3Net [37]	77.8	98.9	99.0	51.2	54.8	97.9	58.9	49.2	73.5
GramNet [25]	73.6	98.8	98.7	51.5	55.3	95.3	55.1	49.4	72.2
DeFake [40]	79.8	98.6	98.6	71.5	78.0	98.4	78.3	74.3	84.7
UnivFD [33]	91.4	96.4	96.1	58.0	73.4	94.5	67.8	57.7	79.4
DIRE [48]	50.4	99.9	99.9	52.3	67.2	99.9	50.1	49.9	71.2
LaRE ² [28]	74.0	99.9	99.9	61.7	88.5	99.9	97.2	68.7	86.2
DRCT [4]	91.5	95.0	94.4	79.4	89.1	94.6	90.0	81.6	89.4
ESSP [5]	82.6	99.2	99.3	78.9	88.9	98.6	96.0	73.9	90.6
CoDE [2]	86.8	99.0	99.2	80.0	92.8	98.8	97.0	94.3	93.5
DLFE [54]	95.5	96.1	96.3	95.9	96.0	96.4	96.0	95.7	96.0
CoD [20]	96.0	99.7	99.8	85.2	95.9	98.2	96.8	98.3	96.2
SDID (ours)	96.9	99.9	99.9	95.9	98.7	99.1	98.0	98.8	98.4

Table 1. Performance comparison on GenImage. Methods are trained on GenImage/SDv1.4 following [56] and evaluated on different testing subsets. The evaluation metric is averaged accuracy with a 0.5 threshold.

Methods	CNNSpot [47]	F3Net [37]	GramNet [25]	DeFake [40]	UnivFD [33]	DIRE [48]	DRCT [4]	DLFE [54]	SDID (ours)
LDM	99.9	99.9	99.4	92.1	98.3	98.2	96.7	97.1	97.6
SDv1.4	99.9	99.8	99.0	99.5	96.2	99.9	96.3	97.1	96.9
SDv1.5	99.9	99.8	98.8	99.5	96.3	99.9	96.3	97.1	96.6
SDv2	97.5	88.7	95.3	89.7	93.8	68.2	94.9	97.1	97.2
SDXL	66.3	55.9	62.6	64.0	91.0	53.8	96.2	97.1	93.7
SDXL-Refiner	86.5	87.4	80.7	69.2	93.9	71.9	93.5	97.1	94.4
SD-Turbo	86.2	68.3	71.2	92.0	86.4	58.9	93.4	97.1	93.2
SDXL-Turbo	72.4	63.7	69.3	93.9	85.9	54.3	92.9	97.1	96.8
LCM-SDv1.5	98.3	97.4	93.1	99.1	90.4	99.8	91.2	97.1	96.3
LCM-SDXL	61.7	55.0	57.0	70.9	89.0	59.7	95.0	97.1	91.3
SDv1-Ctrl	98.0	98.0	90.0	59.0	90.4	99.6	95.6	97.1	97.0
SDv2-Ctrl	85.9	72.4	75.6	62.3	81.1	64.2	92.7	96.8	95.7
SDXL-Ctrl	82.8	82.0	82.7	66.7	89.1	59.1	92.0	94.7	96.6
SDv1-DR	60.9	65.4	51.2	50.1	52.0	52.0	94.1	92.3	86.7
SDv2-DR	51.4	50.4	50.0	50.2	51.0	50.0	69.6	51.8	75.9
SDXL-DR	50.3	50.3	50.1	50.0	50.5	50.0	57.4	49.7	71.9
Avg.	81.1	77.1	76.6	75.5	83.5	71.2	90.5	90.9	92.4

Table 2. Performance comparison on DRCT-2M. Methods are trained on DRCT-2M/SDv1.4 following DRCT [4] and evaluated on different testing subsets. The evaluation metric is averaged accuracy with a 0.5 threshold.

Specifically, SDID performs significantly better than most of the methods on the more challenged Midjourney, ADM, and BigGAN test-sets. In Table 2, we find that the proposed SDID outperforms all other methods by at least 1.5 points. Especially, the SDID performs significantly better than most methods on the diffusion models like SD-v2-DR and SDXL-DR, since they are challenging test sets where images are reconstructed by diffusion models from real images rather than random noise. As shown in Table 3, the proposed SDID outperforms all other methods by at least 9 points. The proposed SDID achieves more than 95% on most test-sets, except for the PG-v2-256, FLUX.1-sch, and FLUX.1-dev test-sets. Overall, the results on Table 1, 2, 3

show the superiority of the proposed SDID.

4.3. Ablation Study

SDID. As shown in Table 4, when concatenating both RGB and NPR features, the detector achieves 90.1% accuracy, which is better than the detector that only uses RGB features (86.7%) or NPR features (88.5%). This shows that both RGB and NPR modalities have advantages in generated image detection. Furthermore, CMCL boosts performance from 90.1% to 95.4%, showing the effectiveness of CMCL. Finally, CMMD improves performance from 95.4% to 98.4%, indicating the effectiveness of CMMD.

Different Inputs. We also validate the effectiveness of

Methods	CNNSpot [47]	FreqFD [14]	LNP [22]	UnivFD [33]	DIRE [48]	FreqNet [43]	NPR [44]	DRCT [4]	CO-SPY [9]	SDID (ours)
LDM	77.6	54.2	84.9	79.1	66.3	74.9	84.3	79.3	95.0	97.5
SD-v1.4	90.0	62.9	95.9	80.9	83.5	69.2	90.9	81.2	92.0	97.6
SD-v1.5	90.0	62.6	96.2	80.9	83.8	68.9	91.3	81.1	91.3	97.6
SSD-1B	66.6	49.7	79.1	76.5	56.6	49.2	47.9	75.8	83.2	96.2
tiny-sd	66.4	52.1	81.5	77.0	63.6	63.6	88.4	80.0	84.8	97.5
SegMoE-SD	74.4	51.7	86.6	83.1	66.0	64.4	93.8	75.1	89.5	97.5
small-sd	70.1	52.6	80.7	77.4	68.4	65.4	89.1	81.2	85.8	97.5
SD-2-1	68.1	49.9	57.2	81.7	65.2	51.6	51.3	76.1	88.5	97.1
SD-3-medium	60.7	50.0	53.7	78.4	56.8	49.6	50.0	75.0	82.9	95.0
SDXL-turbo	88.7	61.3	83.5	84.3	73.0	66.6	83.6	80.4	95.4	96.1
SD-2	65.7	49.8	54.6	73.8	60.1	50.6	51.2	75.1	83.7	96.3
SDXL	61.8	49.7	80.3	63.6	51.9	48.8	47.7	75.2	74.1	95.5
PG-v2.5-1024	53.6	49.7	79.1	78.2	52.3	48.7	47.7	71.3	88.7	94.8
PG-v2-1024	63.5	49.7	52.5	78.5	56.6	48.8	48.4	66.6	89.1	96.4
PG-v2-512	57.9	49.9	49.4	58.9	53.6	49.1	49.1	77.6	64.9	96.0
PG-v2-256	63.3	50.2	54.8	63.0	60.3	49.4	49.7	73.5	72.9	90.5
PAXL-2-1024	56.3	49.8	53.4	80.1	54.8	49.4	51.2	71.8	93.9	96.4
PAXL-2-512	65.4	52.2	74.2	80.3	62.2	57.2	81.3	75.5	94.9	97.3
LCM-sd-xl	81.5	52.4	85.5	78.0	68.6	62.3	50.9	81.1	96.2	97.4
LCM-sdv1-5	92.3	68.2	90.9	79.7	79.0	76.2	93.7	79.6	97.1	97.5
FLUX.1-sch	56.0	50.0	54.4	72.9	56.3	50.4	53.1	68.4	85.2	93.9
FLUX.1-dev	57.4	50.0	54.1	75.9	56.3	49.1	50.7	70.7	86.1	93.6
Avg.	69.4	53.1	71.9	76.5	63.4	57.4	65.7	76.0	87.1	96.1

Table 3. Performance comparison on Co-Spy-Bench. Methods are trained on DRCT-2M/SDv1.4 following CO-SPY [9] and evaluated on different testing subsets. The evaluation metric is averaged accuracy with a 0.5 threshold.

RGB	NPR	CMCL	CMMD	GenImage
✓				86.7
	✓			88.5
✓	✓			90.1
✓	✓	✓		95.4
✓	✓	✓	✓	98.4

Table 4. Ablation study of SDID. All models are trained on the GenImage/SDv1.4 dataset, and the averaged accuracy (%) is reported on all test sets of GenImage.

Input	CMCL	CMMD	GenImage
RGB & RGB			86.9
RGB & RGB	✓		93.0
RGB & RGB	✓	✓	93.2
RGB & High Freq			88.3
RGB & High Freq	✓		93.8
RGB & High Freq	✓	✓	95.5
RGB & NPR			90.1
RGB & NPR	✓		95.4
RGB & NPR	✓	✓	98.4

Table 5. Ablation of different inputs on GenImage. Note that the "RGB & RGB" with CMCL degenerates to the conventional contrastive learning.

the proposed CMCL and CMMD on different inputs. As shown in Table 5, "RGB & RGB" denotes that we change the NPR input of SDID to another RGB input with random augmentation. Note that, in this setting, "RGB & RGB" with CMCL degenerates to the conventional contrastive learning. We also notice that CMMD not works well in "RGB & RGB" setting, since there is only the RGB modality. "RGB & High Freq" means that we change the NPR input to the high frequency component of the image, since there are evidences that the high frequency of the image is useful to detect fake images [15, 27, 43]. Table 5 shows that the proposed CMCL and CMMD can consistently boost the performance of the detector under different inputs, and the NPR modality has advantages in generated

image detection.

CMCL and CMMD. The differences between Table 6 and Table 4 are that only the RGB features are used to predict whether the input image is real or fake. The NPR features are only used to perform CMCL and CMMD. As shown in Table 6, the proposed CMCL boosts the performance from 86.7% to 93.6% by 6.9 points, while the proposed CMMD further boosts the performance from 93.6% to 96.4% by 2.8 points. Note that all experiments in Table 6 only use the RGB encoder to detect generated images

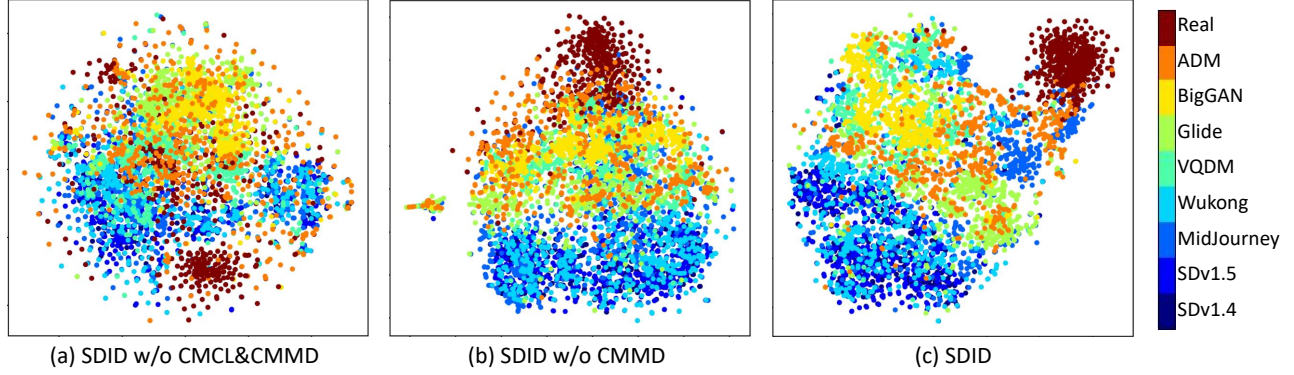


Figure 4. The t-SNE visualization of embeddings. "SDID w/o CMCL&CMMD" denotes not using CMCL and CMMD. The dark red color denotes the real images.

RGB	CMCL	CMMD	GenImage
✓			86.7
✓	✓		93.6
✓	✓	✓	96.4

Table 6. Ablation study of CMCL and CMMD. The differences between this Table and Table 4 are that only the RGB features are used to predict whether the input image is real or fake. The NPR features are only used to perform CMCL and CMMD. All models are trained on the GenImage/SDv1.4 dataset, and the averaged accuracy (%) is reported on all test sets of GenImage.

RGB	NPR	CMCL	CMMD	GenImage
✓				81.3
	✓			88.5
✓	✓			89.3
✓	✓	✓		93.2
✓	✓	✓	✓	95.5

Table 7. Generalized to different RGB encoder. The differences between this Table and Table 4 are that the RGB encoder is changed to the CLIP image encoder. All models are trained on the GenImage/SDv1.4 dataset, and the averaged accuracy (%) is reported on all test sets of GenImage.

without any new model architecture involved. The proposed CMCL and CMMD with NPR modality can boost the performance of vanilla RGB encoder by 9.7 points without increased time cost during testing.

We also apply t-SNE [46] to show the embedding distribution among "SDID w/o CMCL&CMMD", "SDID w/o CMMD", and "SDID" on the GenImage benchmark. From Figure 4 (a) and Figure 4 (b), we can see that the embeddings extracted from "SDID w/o CMMD" have better inter-class separability compared with "SDID w/o CMCL&CMMD", further indicating that CMCL can learn more discriminative features. Figure 4 (c) and Figure 4 (b) show that the embeddings of real images extracted from "SDID" have better intra-class compactness compared with "SDID w/o CMMD", further indicating that CMMD is capable to model the relationships within the real images.

Generalized to different RGB encoders. To validate that the proposed method can be generalized to different RGB encoders, we change the RGB encoder from DINOv2 to the CLIP image encoder in Table 7. The Table 7 shows the proposed method could still boost the performance when using the CLIP image encoder as the RGB encoder. For example, the CMCL boosts the performance from 89.3% to 93.2% by 3.9 points, and the CMMD boosts

the performance from 93.2% to 95.5% by 2.3 points.

5. Conclusion

In this paper, to optimize the embedding space tailored for fake image detection, we leverage the RGB and NPR modalities to learn forgery-aware embedding space via two proposed representation learning methods, Cross-Modal Contrastive Learning (CMCL) and Cross-Modal Mutual Distillation (CMMD). CMCL aims at boosting the discrimination of features between real and fake images by contrastive learning. CMMD transfers the learned knowledge between two modalities, which is capable to learn compact features within the intra-class (real class or fake class). The proposed CMCL and CMMD are complementary and work collaboratively so that each modality learns a more comprehensive forgery-aware representation to distinguish real and fake images. The proposed method is evaluated on various benchmarks with extensive experiments, and achieves state-of-the-art results on all of them.

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